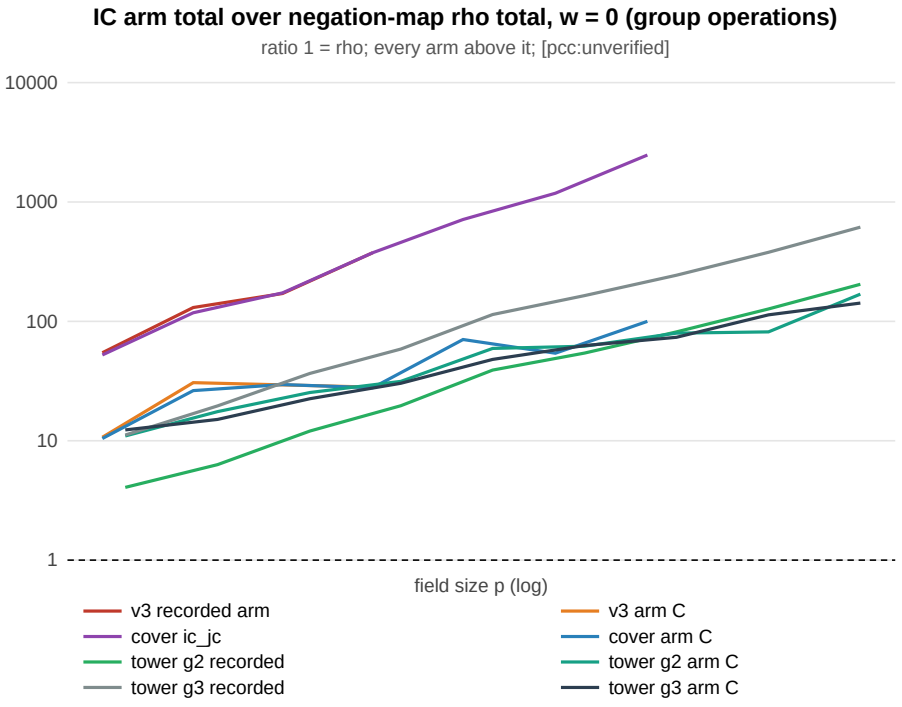
Negation re-measurement

Negation-map re-measurement of v3, tower_large and cover_pushforward: results (2026-11-13)



figure

Protocol: PROTOCOL.md, hashed in frozen.sha256 (8771b2dca99318cd058a9c19ddc1a92d8ac7bef68aeeed630e06c2bda53cba86) before the first run. Code: `src/cryptanalysis/chained_transfer/negation_remeasure.rs`, `src/bin/negation_remeasure.rs` (plus one `mod.rs` line). Data: remeasure.json (every row), remeasure.svg, run.log, the parts `remeasure_v3.json`, `remeasure_cover.json`, `remeasure_tower*.json`; derived files tables.md (every arm-row), summary.json, scorecard.json and the scripts that make them (`render_tables.py`, `scorecard.py`, `condensed.py`, `assemble_results.py`, `add_metrics.py`). Bench record (left open): `benchmarks/records/20261113T0320Z-negation-remeasure.json`. Every number in this file is a `[pcc:unverified]` assertion: no independent validator has replayed it, and no pcc claim was filed.

Answer

The three targets now have what the validation asked for: a negation-map rho measured in the same process on the same curve and instance, with its canonicalisations, partition hashes, DP-table lookups and inserts (and the walk's word operations) emitted per run, and every arm's emitted native counters priced the same way at the 5a band and at a host-calibrated weight [pcc:unverified]. All 162 arm-rows in 3 targets were classified. The classification result:

- **No arm is below the matched rho at all three prices** (0 rows below). Every improved arm (A, B, C) and every recorded arm is `above` the negation rho in the band, except the cover arms that stop at the first collision and emit no counter at all, which are `above` or `unresolved (lower bound)` [pcc:unverified].
- **Distances grow when the reference is corrected.** The recorded plain `rho_reference` spent about 400 operations on set-up where the tuned negation walk spends 50 to 80, so the recorded distances were too small. At the largest prime of each target the recorded arm's distance (0 / 0.1 / 1 GAE) moves from 40.8 / 112 / 754 to 376 / 864 / 2,353 (v3, p = 1619), from 394 / 4,720 / 43,658 to 2,473 / 20,667 / 51,693 (cover ic_jc, p = 12841) and from 142 / 159,687 / 1,595,592 to 616 / 446,318 / 995,132 (tower genus 3, p = 64007) [pcc:unverified]. This is an accounting change, not an improvement.
- **Recorded verdicts that change** (decision rule of the protocol): tower genus 2 at p = 251, `unresolved` to `above`; and in cover_pushforward the three arms that stop at the first collision (J(C), E1 fibre, birthday control) were recorded `below` the plain rho at p = 211 to 3203 (15 rows, ratios 0.17 to 0.85); against the negation rho they are `above` (or `unresolved (lower bound)` for the birthday control at p = 211 to 1619 and for all three at p = 211): the recorded `below` was the start-up of the plain rho, not a property of the cover [pcc:unverified]. The other 48 recorded-arm rows keep their class.
- **The improved arms are a measured improvement of the index-calculus baseline** (N7): arm C uses 2,583 group operations at v3 p = 1619 against 34,521 recorded, and its priced totals 2,583 / 23,440 / 211,155 are below the recorded arm's lower-bound totals 34,521 / 94,893 / 638,242 at all three prices; the same holds for tower genus 3 at p = 16001 and for cover J(C) at p = 12841 (`scorecard.json` N7). The gap to the matched rho remains: arm C is 28.1 / 213 / 778 times the negation rho at v3 p = 1619 (0 / 0.1 / 1 GAE) [pcc:unverified].
- **Calibrated reading.** `w_cal` (best-of-N, unisolated VM, see Method) is 0.0068 to 0.0079 for a mod-p multiplication on the word genus-2 Jacobian and 0.0041 to 0.0044 on the genus-3 one; rho's canonicalisation and partition hash cost 0.011 to 0.016 and 0.010 to 0.015 of an elliptic addition. The calibrated order (0, `w_cal`, 10 `w_cal`) agrees with the band verdict for every improved arm and every recorded arm with an emitted counter; it differs only for the first-collision arms whose counters are not emitted (band `unresolved (lower bound)`, calibrated `above`) [pcc:unverified].

Readiness for the validation gate (the task's list), per target:

target	negation rho in the same process	rho native counters emitted	every IC arm's counters listed (gaps flagged)	every log certified	`w_cal` with method and host	one-time transfer apart
v3	yes (32 seeds per prime)	yes	yes; recorded collector is a lower bound	104/104 IC, 256/256 rho (and the recorded rho and IC)	yes	yes
tower_large	yes (64 seeds x 3 instances	yes	yes; recorded `ic::solve`	1020/1020 IC,	yes	yes

target	negation rho in the same process	rho native counters emitted	every IC arm's counters listed (gaps flagged)	every log certified	`w_cal` with method and host	one-time transfer apart
	per prime)		counters all emitted	3456/3456 rho		
cover_pushforward	yes (32 seeds per prime)	yes	yes; E1 arms and J(C) collector are lower bounds	315/315 IC, 448/448 rho	yes	yes

The binary exited 0 only when every log (IC and rho, recorded and new, and the in-process recorded plain rho) passed `[k]G == T`, and the reproduction gates passed at every row (v3: the recorded candidate, rho ops, IC ops, trials and LA counter of all four primes; tower: the instance (r, #E, #J) of every instance and, where the seed counts agree, the recorded rho and IC means; cover: the recorded group ops of four arms for every seed). Admission to the gate is the validator's call, not this run's [pcc:unverified].

Method (and where it departs from PROTOCOL.md)

- One process per section, 4 threads, host `Intel(R) Xeon(R) Processor @ 2.10GHz`, 4 vCPU, a shared VM (other agents compiled and ran tests during the runs; `run.log` records the load average at each launch: 1.9 to 5.9). It ran locally under the bench record, as the earlier instrument rerun did, not on isolab-eks, and no S3 resume checkpoint was possible (no upload token in this container).
- Matched rho: `rho_reference_walk`, tuned negation walk (primary) and plain walk, seeds as in the protocol, on the same `E/E1` as the IC arms; the recorded plain `rho_reference`/ `rho_on_ec` is run in the same process as the "old" column. rho counts every group operation (set-up, starts, walk, verification). Native counters emitted per run: `canonicalisations` (the walk's counter), `partition_hashes` (derived exactly from a `key`-counting wrapper: all key evaluations minus the two per canonicalisation minus one per table lookup), `dp_lookups` (= distinguished points), `dp_inserts` (= lookups minus table hits), `word_ops` (`work_native` of the walk: the short-cycle window compares). `negation_remeasure::tests` checks that the wrapper leaves the walk unchanged (totals equal `instrument::rho_on_curve`) and that the counters add up.
- **Deviation 1: word operations are priced.** PROTOCOL.md listed four rho classes (canon, hash, lookup, insert) in the pricing paragraph; the code also prices the walk's `work_native` word operations (one class, `word`), because AGENTS.md 5a prices every emitted counter. This raises rho's priced total and so lowers every IC/rho ratio (it works against the finding that IC is above rho). N2 and N3 are scored on the four named classes and, for N3, also as measured.
- **Deviation 2: the 5a calibration uses 25 batches of 20,000 operations for every class and every group addition** (as frozen) but `inv_r` uses batches of 1,000 (a Fermat inversion is about 40 multiplications); the class `draw` is a xorshift step; the table classes use the walk's own `MixHasher` table (512 entries). `w_cal` = class time over the time of one addition of the arm's own group (E for rho; the word Jacobian for arms A, B, C; the recorded `Genus2Curve` implementation for the recorded v3/cover collectors; the genus-2 or genus-3 word Jacobian per rung for the tower). It is a best-of-N figure on an unisolated host, not a `perf-ipc` measurement [pcc:unverified].
- **Deviation 3: tower p = 64007 used 4 algorithm seeds for the recorded arm and 2 for A, B, C** (frozen text: 8 and 4; all other primes used 8 and 4). The reproduction gate at 64007 therefore checks the instances and the 64 old rho seeds only. All three instances completed at p = 32003 and p = 64007.
- **Deviation 4: the cover recorded arms use the original 3 seeds** and the matched rho, the improved arms and the aut check use the seed-0 instance of each prime (rho 32 seeds, A, B, C 8 seeds); recorded distances are the original per-seed median ratios.

- **Deviation 5: the PROTOCOL.md prose has 5 lint-prose findings** (the words “improved” and “improvement” in paragraphs without a tag); the file is frozen by hash and was not edited. The files written after the freeze lint at 0.
- **Deviation 6: three binary builds.** The measured parts (`remeasure_v3`, `_cover`, `_tower`, `_tower_p32003`, `_tower_p64007`) came from the first build of `negation_remeasure`; after them the binary was reformatted (`rustfmt`) and a per-instance `curves_checked` field and the merge of `_aut` parts were added, which touch no measured value. The `_aut` part (tower with all seed counts 0, no calibration: it builds every tower instance up to p = 64007 and lists its `j`-invariant) and the final `merge` used the rebuilt binary. The bench record binds the dirty working tree of branch `claude/ecdlp-chained-transfers-8cpiek` at origin/main `47dc95a934`; the code is not committed in this session (the validation’s wish for a record bound to a clean commit containing the measuring code needs the PR commit and a re-run or an amendment) [pcc:unverified].
- Automorphism fold: 38 curves checked (v3 4, cover 7, tower instances at p = 251 to 64007); 0 has `j` in {0, 1728}. The fold is therefore vacuous on this data and was not implemented for the `chained_transfer::ec::Curve` type [pcc:unverified].

Matched rho: group operations and native counters (mean over seeds)

v3

p	r	neg total	neg walk	canon	part. hash	DP lookup	DP insert	word ops	native / step (all)	native / step (4 classes)	plain tota
211	109	62.7	12.0	16.2	16.2	4.09	3.09	31.6	7.1	3.96	68.3
419	113	64.9	13.6	18.6	18.6	4.81	3.75	35.8	7.26	4.08	71.6
821	71.0	66.4	10.5	15.7	15.7	5	3.97	23.6	7.2	4.55	69.6
1619	401	91.9	26.2	31.2	31.2	4.91	3.84	108	8.18	3.25	97.4

cover_pushforward

p	r	neg total	neg walk	canon	part. hash	DP lookup	DP insert	word ops	native / step (all)	native / step (4 classes)	pl to
211	109	64.2	13.5	18.0	18.0	4.47	3.47	38.4	7.11	3.8	7
419	113	75.8	24.8	30.1	30.1	5.06	4.06	72.9	7.52	3.67	6
821	71.0	66.0	11.2	16.2	16.2	4.84	3.81	26.0	7.19	4.4	6
1619	401	92.9	27.1	32.5	32.5	5.25	4.25	106	7.98	3.3	1
3203	313	84.2	22.3	27.5	27.5	5.09	4.09	83.0	7.67	3.35	8
6421	3,191	159	78.1	87.2	87.2	9	8	446	9.87	2.96	1
12841	3,169	151	75.0	84.1	84.1	9.03	8.03	460	10.1	2.89	1

tower_large

p	r	neg total	neg walk	canon	part. hash	DP lookup	DP insert	word ops	native / step (all)	native / step (4 classes)	p
251	107	68.6	12.5	18.3	18.3	5.65	4.64	29.3	7.3	4.5	
503	260	84.2	24.8	29.7	29.7	4.67	3.66	108	8.28	3.19	
1009	409	85.7	24.5	30.1	30.1	5.48	4.47	93.6	7.89	3.38	
2003	1,010	103	36.3	44.3	44.3	8.01	7.01	145	8.02	3.34	
4001	1,018	104	38.3	46.1	46.1	7.72	6.72	159	8.13	3.26	
8009	3,322	148	74.8	83.6	83.6	8.73	7.72	475	10.4	2.89	
16001	6,672	197	118	129	129	11.0	9.95	813	11.0	2.8	
32003	13,395	252	165	182	182	16.5	15.5	1,161	11.0	2.79	
64007	26,650	313	215	237	237	21.6	20.5	1,458	10.8	2.82	

The old plain rho column is the recorded `rho_reference` (v3: the single recorded seed; cover: the three original seeds; tower: 64 seeds). The negation total is `4.3` to `7.2` times smaller on the tower, `5.2` to `8.0` on cover and `5.5` to `9.2` on v3. The negation walk is not always cheaper than the plain tuned walk on a small `r` (cover p = 419 and others, a few seeds' difference): the table shows both and the `best of negation and plain` columns of `tables.md` use the cheaper at each price [pcc:unverified].

Calibration (best-of-N timing, ns per operation) and `w_cal`

target	p	mul_p	mul_r	inv_r	canon	hash	lookup	insert	draw
cover_pushforward	211	6.21	6.21	64.2	2.17	2.07	4.2	7.32	2.07
cover_pushforward	419	6.21	6.21	66.6	2.28	2.07	4.18	7.26	2.07
cover_pushforward	821	6.21	6.21	59.3	2.19	2.07	4.17	7.22	2.07
cover_pushforward	1619	6.21	6.21	79.0	2.18	2.07	4.21	7.26	2.07
cover_pushforward	3203	6.21	6.21	79.0	2.17	2.07	4.2	7.24	2.07

target	p	mul_p	mul_r	inv_r	canon	hash	lookup	insert	draw
cover_pushforward	6421	6.21	6.21	100	2.18	2.07	4.2	7.33	2.07
cover_pushforward	12841	6	6.21	102	2.38	2.07	4.19	7.24	2.07
tower_large	251	6.21	6.21	66.6	2.28	2.07	4.2	7.27	2.07
tower_large	503	6.21	6.21	74.2	2.27	2.07	4.21	7.22	2.07
tower_large	1009	6.21	6.21	77.6	2.19	2.07	4.19	7.37	2.07
tower_large	2003	6.21	6.21	92.4	2.19	2.07	4.19	7.2	2.07
tower_large	4001	6.21	6.21	90.0	2.18	2.07	4.17	7.2	2.07
tower_large	8009	6.21	6.21	101	2.17	2.07	4.19	7.27	2.07
tower_large	16001	6.21	6.21	100	2.18	2.07	4.19	7.35	2.07
tower_large	32003	6.21	6.21	113	2.19	2.07	4.21	7.2	2.07
tower_large	64007	6.21	6.21	126	2.29	2.07	4.19	7.26	2.07
v3	211	6.21	6.21	64.1	2.26	2.07	4.19	7.36	2.07
v3	419	6.21	6.21	66.6	2.26	2.07	4.17	7.34	2.07
v3	821	6.21	6.21	59.3	2.18	2.07	4.2	7.31	2.07
v3	1619	6.21	6.21	79.0	2.18	2.07	4.2	7.3	2.07

Recorded arms and improved arms, old versus new distance

Old = the recorded ratio to the recorded plain `rho_reference` (group ops plus the emitted LA counter at the price, as in the original files). New = the arm's total, every emitted native counter priced at the same weight, over the negation-map rho total with its native counters priced at that weight. `cal.` is the calibrated reading. Lower-bound arms have unemitted counters and are never classed `below`. [pcc:unverified]

cover_pushforward: recorded arms, old (recorded ratio to the plain `rho_reference`) versus new (to the negation-map rho)

p	arm	group ops	old ratio 0 / 0.1 / 1	old class	new ratio 0 / 0.1 / 1	new class (band)	cal. ratio 0 / w_cal / 10 w_cal
211	ic_jc	3,359	7.61 / 10.9 / 40.8	above	52.4 / 66.6 / 123	above	52.4 / 51.9 / 48.3
211	ic_e1_two_term	762	1.73 / 2.63 / 10.4	above	11.9 / 16.1 / 32.8	above	11.9 / 14.4 / 33.4
211	ic_e1_fibre_full	3,359	7.61 / 8.48 / 16.3	above	52.4 / 51.7 / 49.2	above	52.4 / 54.0 / 66.8
211	ic_jc_first_collision	131	0.338 / 0.338 / 0.338	below	2.05 / 1.81 / 0.896	unresolved (lower bound)	2.05 / 2.01 / 1.74
211	ic_e1_fibre_first_collision	131	0.338 / 0.338 / 0.338	below	2.05 / 1.81 / 0.896	unresolved (lower bound)	2.05 / 2.01 / 1.74
211	birthday_e1_first_collision	131	0.338 / 0.338 / 0.338	below	2.05 / 1.81 / 0.896	unresolved (lower bound)	2.05 / 2.01 / 1.74
419	ic_jc	8,932	23.7 / 37.7 / 157	above	118 / 157 / 279	above	118 / 117 / 107
419	ic_e1_two_term	673	1.73 / 2.43 / 8.79	above	8.89 / 11.0 / 17.8	above	8.89 / 10.4 / 21.5
419	ic_e1_fibre_full	8,932	23.7 / 25.4 / 39.2	above	118 / 106 / 67.5	above	118 / 118 / 122
419	ic_jc_first_collision	296	0.77 / 0.77 / 0.77	below	3.91 / 3.29 / 1.36	above	3.91 / 3.82 / 3.18
419	ic_e1_fibre_first_collision	296	0.77 / 0.77 / 0.77	below	3.91 / 3.29 / 1.36	above	3.91 / 3.82 / 3.18
419	birthday_e1_first_collision	178	0.35 / 0.35 / 0.35	below	2.34 / 1.97 / 0.815	unresolved (lower bound)	2.34 / 2.29 / 1.91

p	arm	group ops	old ratio 0 / 0.1 / 1	old class	new ratio 0 / 0.1 / 1	new class (band)	cal. ratio 0 / w_cal / 10 w_cal
821	ic_jc	11,426	30.5 / 73.8 / 467	above	173 / 384 / 1,323	above	173 / 175 / 192
821	ic_e1_two_term	467	1.24 / 1.68 / 5.55	above	7.08 / 8.79 / 16.4	above	7.08 / 7.98 / 15.1
821	ic_e1_fibre_full	11,426	30.5 / 32.5 / 50.9	above	173 / 168 / 144	above	173 / 175 / 193
821	ic_jc_first_collision	163	0.398 / 0.398 / 0.398	below	2.47 / 2.24 / 1.22	above	2.47 / 2.44 / 2.17
821	ic_e1_fibre_first_collision	163	0.398 / 0.398 / 0.398	below	2.47 / 2.24 / 1.22	above	2.47 / 2.44 / 2.17
821	birthday_e1_first_collision	118	0.308 / 0.308 / 0.308	below	1.78 / 1.62 / 0.883	unresolved (lower bound)	1.78 / 1.76 / 1.57
1619	ic_jc	34,876	70.5 / 195 / 1,315	above	375 / 858 / 2,337	above	375 / 379 / 409
1619	ic_e1_two_term	2,755	5.68 / 16.5 / 114	above	29.6 / 70.5 / 196	above	29.6 / 49.2 / 196
1619	ic_e1_fibre_full	34,876	70.5 / 85.8 / 223	above	375 / 381 / 399	above	375 / 397 / 564
1619	ic_jc_first_collision	567	0.167 / 0.167 / 0.167	below	6.1 / 5.11 / 2.08	above	6.1 / 5.99 / 5.1
1619	ic_e1_fibre_first_collision	567	0.167 / 0.167 / 0.167	below	6.1 / 5.11 / 2.08	above	6.1 / 5.99 / 5.1
1619	birthday_e1_first_collision	227	0.167 / 0.167 / 0.167	below	2.44 / 2.05 / 0.831	unresolved (lower bound)	2.44 / 2.4 / 2.04
3203	ic_jc	60,063	97.1 / 483 / 3,955	above	713 / 3,000 / 10,492	above	713 / 749 / 1,023
3203	ic_e1_two_term	2,203	3.62 / 8.48 / 56.1	above	26.1 / 56.6 / 156	above	26.1 / 40.3 / 148
3203	ic_e1_fibre_full	60,063	97.1 / 117 / 301	above	713 / 733 / 799	above	713 / 754 / 1,068
3203	ic_jc_first_collision	476	0.853 / 0.853 / 0.853	below	5.65 / 4.81 / 2.06	above	5.65 / 5.55 / 4.8

p	arm	group ops	old ratio 0 / 0.1 / 1	old class	new ratio 0 / 0.1 / 1	new class (band)	cal. ratio 0 / w_cal / 10 w_cal
3203	ic_e1_fibre_first_collision	476	0.853 / 0.853 / 0.853	below	5.65 / 4.81 / 2.06	above	5.65 / 5.55 / 4.8
3203	birthday_e1_first_collision	324	0.473 / 0.473 / 0.473	below	3.84 / 3.27 / 1.4	above	3.84 / 3.77 / 3.26
6421	ic_jc	188,083	198 / 1,226 / 10,476	above	1,185 / 5,295 / 12,660	above	1,185 / 1,249 / 1,681
6421	ic_e1_two_term	28,508	29.5 / 447 / 4,201	above	180 / 1,873 / 4,907	above	180 / 987 / 6,474
6421	ic_e1_fibre_full	188,083	198 / 452 / 2,736	above	1,185 / 1,949 / 3,317	above	1,185 / 1,662 / 4,904
6421	ic_jc_first_collision	1,750	1.39 / 1.39 / 1.39	above	11.0 / 7.87 / 2.2	above	11.0 / 10.7 / 8.33
6421	ic_e1_fibre_first_collision	1,750	1.39 / 1.39 / 1.39	above	11.0 / 7.87 / 2.2	above	11.0 / 10.7 / 8.33
6421	birthday_e1_first_collision	1,463	1.05 / 1.05 / 1.05	above	9.22 / 6.58 / 1.84	above	9.22 / 8.93 / 6.96
12841	ic_jc	372,747	394 / 4,720 / 43,658	above	2,473 / 20,667 / 51,693	above	2,473 / 2,819 / 5,163
12841	ic_e1_two_term	28,904	30.4 / 486 / 4,585	above	192 / 2,251 / 5,763	above	192 / 1,143 / 7,574
12841	ic_e1_fibre_full	372,747	394 / 940 / 5,858	above	2,473 / 4,125 / 6,941	above	2,473 / 3,475 / 10,257
12841	ic_jc_first_collision	1,963	1.85 / 1.85 / 1.85	above	13.0 / 9.12 / 2.47	above	13.0 / 12.6 / 9.79
12841	ic_e1_fibre_first_collision	1,963	1.85 / 1.85 / 1.85	above	13.0 / 9.12 / 2.47	above	13.0 / 12.6 / 9.79
12841	birthday_e1_first_collision	1,963	1.85 / 1.85 / 1.85	above	13.0 / 9.12 / 2.47	above	13.0 / 12.6 / 9.79

cover_pushforward: improved arms against the negation-map rho

p	arm	group ops	native (by class)	ratio 0 / 0.1 / 1	class	cal. ratio	class (cal)
211	A	824	mul_p 36,089, mul_r 34,750	12.8 / 109 / 489	above	12.8 / 21.2 / 85.1	above
211	B	762	mul_p 30,841, mul_r 11,125	11.9 / 68.5 / 292	above	11.9 / 16.8 / 54.1	above
211	C	670	mul_p 25,174, mul_r 1,501	10.5 / 46.1 / 187	above	10.5 / 13.5 / 36.8	above
419	A	1,752	mul_p 86,433, mul_r 103,935	23.1 / 231 / 881	above	23.1 / 42.1 / 181	above
419	B	1,553	mul_p 73,049, mul_r 3,925	20.5 / 103 / 360	above	20.5 / 27.9 / 82.2	above
419	C	1,993	mul_p 94,729, mul_r 1,276	26.3 / 129 / 450	above	26.3 / 35.5 / 103	above
821	A	2,186	mul_p 156,164, mul_r 134,447	33.1 / 430 / 2,198	above	33.1 / 66.7 / 332	above
821	B	1,407	mul_p 92,024, mul_r 8,254	21.3 / 157 / 763	above	21.3 / 32.8 / 123	above
821	C	1,953	mul_p 127,933, mul_r 1,028	29.6 / 204 / 983	above	29.6 / 44.3 / 161	above
1619	A	3,658	mul_p 334,834, mul_r 1,197,399	39.4 / 1,414 / 5,621	above	39.4 / 162 / 1,081	above
1619	B	2,993	mul_p 251,513, mul_r 27,761	32.2 / 279 / 1,033	above	32.2 / 54.0 / 218	above
1619	C	2,583	mul_p 207,187, mul_r 1,385	27.8 / 211 / 773	above	27.8 / 44.0 / 166	above
3203	A	7,325	mul_p 680,173, mul_r 3,131,700	86.9 / 3,925 / 16,500	above	86.9 / 418 / 2,948	above

p	arm	group ops	native (by class)	ratio 0 / 0.1 / 1	class	cal. ratio	class (cal)
3203	B	4,500	mul_p 381,996, mul_r 38,780	53.4 / 471 / 1,837	above	53.4 / 89.2 / 363	above
3203	C	5,932	mul_p 493,623, mul_r 1,899	70.4 / 561 / 2,166	above	70.4 / 112 / 433	above
6421	A	10,634	mul_p 1,544,468, mul_r 23,298,106	67.0 / 11,215 / 31,198	above	67.0 / 1,185 / 8,788	above
6421	B	7,064	mul_p 938,285, mul_r 87,711	44.5 / 493 / 1,297	above	44.5 / 89.4 / 394	above
6421	C	8,592	mul_p 1,118,677, mul_r 2,948	54.2 / 543 / 1,419	above	54.2 / 103 / 435	above
12841	A	21,489	mul_p 3,355,584, mul_r 59,144,251	143 / 29,137 / 78,581	above	143 / 3,176 / 23,691	above
12841	B	13,945	mul_p 2,004,799, mul_r 115,520	92.5 / 1,050 / 2,683	above	92.5 / 190 / 846	above
12841	C	15,063	mul_p 2,142,707, mul_r 3,627	99.9 / 1,067 / 2,717	above	99.9 / 198 / 859	above

tower_large: recorded arms, old (recorded ratio to the plain `rho_reference`) versus new (to the negation-map rho)

p	arm	group ops	old ratio 0 / 0.1 / 1	old class	new ratio 0 / 0.1 / 1	new class (band)	cal. ratio 0 / w_cal / 10 w_cal	clas (cal
251	g2_recorded	279	0.67 / 8.77 / 81.7	unresolved	4.06 / 72.7 / 365	above	4.06 / 9.65 / 52.8	abov
251	g3_recorded	767	1.84 / 22.8 / 211	above	11.2 / 436 / 2,246	above	11.2 / 31.1 / 185	abov
503	g2_recorded	532	1.02 / 20.5 / 196	above	6.32 / 146 / 553	above	6.32 / 18.5 / 109	abov
503	g3_recorded	1,648	3.15 / 35.3 / 324	above	19.6 / 747 / 2,868	above	19.6 / 56.0 / 326	abov

p	arm	group ops	old ratio 0 / 0.1 / 1	old class	new ratio 0 / 0.1 / 1	new class (band)	cal. ratio 0 / w_cal / 10 w_cal	clas (cal
1009	g2_recorded	1,033	1.67 / 40.9 / 394	above	12.1 / 323 / 1,286	above	12.1 / 38.6 / 239	abov
1009	g3_recorded	3,137	5.08 / 55.4 / 508	above	36.6 / 1,465 / 5,883	above	36.6 / 107 / 641	abov
2003	g2_recorded	2,024	2.84 / 75.7 / 731	above	19.7 / 569 / 2,015	above	19.7 / 68.0 / 417	abov
2003	g3_recorded	6,025	8.44 / 114 / 1,065	above	58.6 / 2,555 / 9,130	above	58.6 / 186 / 1,109	abov
4001	g2_recorded	4,048	5.62 / 147 / 1,419	above	39.1 / 1,099 / 3,776	above	39.1 / 131 / 802	abov
4001	g3_recorded	11,815	16.4 / 346 / 3,317	above	114 / 5,801 / 20,160	above	114 / 406 / 2,534	abov
8009	g2_recorded	8,054	8.4 / 212 / 2,046	above	54.3 / 1,323 / 3,422	above	54.3 / 180 / 1,026	abov
8009	g3_recorded	24,401	25.5 / 993 / 9,698	above	164 / 9,705 / 25,493	above	164 / 722 / 4,477	abov
16001	g2_recorded	16,048	15.0 / 375 / 3,613	above	81.6 / 1,857 / 4,295	above	81.6 / 268 / 1,487	abov
16001	g3_recorded	47,918	44.9 / 4,616 / 45,755	above	244 / 23,688 / 55,887	above	244 / 1,694 / 11,152	abov
32003	g2_recorded	32,004	26.2 / 656 / 6,328	above	127 / 2,840 / 6,243	above	127 / 415 / 2,224	abov
32003	g3_recorded	95,572	78.4 / 26,097 / 260,265	above	379 / 89,891 / 202,177	above	379 / 6,027 / 41,428	abov
64007	g2_recorded	63,987	47.1 / 1,175 / 11,322	above	205 / 4,640 / 10,101	above	205 / 712 / 3,879	abov
64007	g3_recorded	192,688	142 / 159,687 / 1,595,592	above	616 / 446,318 / 995,132	above	616 / 31,305 / 222,705	abov

tower_large: improved arms against the negation-map rho

p	arm	group ops	native (by class)	ratio 0 / 0.1 / 1	class	cal. ratio	class (cal)
251	g2_C	752	mul_p 34,605,	11.0 / 57.0 /	above	11.0 / 14.6 /	above

p	arm	group ops	native (by class)	ratio 0 / 0.1 / 1	class	cal. ratio	class (cal)
			mul_r 1,300	253		42.7	
251	g3_C	845	mul_p 81,954, mul_r 10,031	12.3 / 132 / 642	above	12.3 / 17.8 / 60.2	above
503	g2_C	1,474	mul_p 98,223, mul_r 2,287	17.5 / 113 / 392	above	17.5 / 25.7 / 86.6	above
503	g3_C	1,270	mul_p 176,649, mul_r 15,279	15.1 / 201 / 743	above	15.1 / 24.3 / 92.3	above
1009	g2_C	2,171	mul_p 170,572, mul_r 1,900	25.3 / 190 / 700	above	25.3 / 39.2 / 144	above
1009	g3_C	1,929	mul_p 335,682, mul_r 21,713	22.5 / 369 / 1,441	above	22.5 / 39.6 / 168	above
2003	g2_C	3,235	mul_p 339,082, mul_r 1,658	31.4 / 292 / 979	above	31.4 / 54.0 / 217	above
2003	g3_C	3,114	mul_p 711,593, mul_r 40,982	30.3 / 614 / 2,150	above	30.3 / 59.9 / 273	above
4001	g2_C	6,154	mul_p 674,210, mul_r 2,271	59.4 / 567 / 1,848	above	59.4 / 103 / 419	above
4001	g3_C	4,978	mul_p 1,311,507, mul_r 54,182	48.0 / 1,087 / 3,710	above	48.0 / 101 / 487	above
8009	g2_C	9,192	mul_p 1,299,770, mul_r 2,890	61.9 / 651 / 1,625	above	61.9 / 119 / 507	above
8009	g3_C	9,315	mul_p 3,093,659, mul_r 122,436	62.7 / 1,544 / 3,995	above	62.7 / 149 / 727	above
16001	g2_C	15,684	mul_p 2,579,985, mul_r 5,358	79.8 / 897 / 2,019	above	79.8 / 165 / 719	above
16001	g3_C	14,457	mul_p 5,528,664, mul_r 161,955	73.5 / 1,909 / 4,429	above	73.5 / 186 / 921	above
32003	g2_C	20,587	mul_p 3,926,854, mul_r 6,078	81.7 / 1,015 / 2,186	above	81.7 / 179 / 792	above
32003	g3_C	28,619	mul_p 12,438,794, mul_r 312,484	114 / 3,198 / 7,067	above	114 / 306 / 1,515	above

p	arm	group ops	native (by class)	ratio 0 / 0.1 / 1	class	cal. ratio	class (cal)
64007	g2_C	52,898	mul_p 11,694,102, mul_r 9,178	169 / 2,398 / 5,143	above	169 / 420 / 1,987	above
64007	g3_C	44,494	mul_p 23,093,987, mul_r 429,917	142 / 4,699 / 10,311	above	142 / 454 / 2,397	above

v3: recorded arms, old (recorded ratio to the plain `rho_reference`) versus new (to the negation-map rho)

p	arm	group ops	old ratio 0 / 0.1 / 1	old class	new ratio 0 / 0.1 / 1	new class (band)	cal. ratio 0 / w_cal / 10 w_cal	class (cal)	lower bound
211	recorded	3,419	7.99 / 11.4 / 42.2	above	54.6 / 70.0 / 135	above	54.6 / 54.2 / 50.9	above	yes
419	recorded	8,475	23.7 / 38.3 / 169	above	131 / 187 / 413	above	131 / 130 / 124	above	yes
821	recorded	11,353	30.7 / 75.2 / 476	above	171 / 382 / 1,352	above	171 / 173 / 191	above	yes
1619	recorded	34,521	40.8 / 112 / 754	above	376 / 864 / 2,353	above	376 / 380 / 411	above	yes

v3: improved arms against the negation-map rho

p	arm	group ops	native (by class)	ratio 0 / 0.1 / 1	class	cal. ratio	class (cal)
211	L	3,419	mul_r 14,993	54.6 / 70.5 / 137	above	54.6 / 54.2 / 51.0	above
211	A	824	mul_p 36,089, mul_r 34,750	13.2 / 113 / 535	above	13.2 / 20.9 / 80.9	above
211	B	762	mul_p 30,841, mul_r 11,125	12.2 / 71.1 / 319	above	12.2 / 16.7 / 51.7	above
211	C	670	mul_p 25,174, mul_r 1,501	10.7 / 47.8 / 204	above	10.7 / 13.5 / 35.4	above
419	L	8,475	mul_r 52,288	131 / 187 / 415	above	131 / 130 / 125	above

p	arm	group ops	native (by class)	ratio 0 / 0.1 / 1	class	cal. ratio	class (cal)
419	A	1,752	mul_p 86,433, mul_r 103,935	27.0 / 284 / 1,312	above	27.0 / 47.1 / 202	above
419	B	1,553	mul_p 73,049, mul_r 3,925	23.9 / 127 / 536	above	23.9 / 31.8 / 92.7	above
419	C	1,993	mul_p 94,729, mul_r 1,276	30.7 / 159 / 669	above	30.7 / 40.6 / 116	above
821	L	11,353	mul_r 165,008	171 / 382 / 1,353	above	171 / 173 / 191	above
821	A	2,186	mul_p 156,164, mul_r 134,447	32.9 / 429 / 2,246	above	32.9 / 62.9 / 301	above
821	B	1,407	mul_p 92,024, mul_r 8,254	21.2 / 157 / 780	above	21.2 / 31.4 / 113	above
821	C	1,953	mul_p 127,933, mul_r 1,028	29.4 / 204 / 1,004	above	29.4 / 42.5 / 147	above
1619	L	34,521	mul_r 605,131	376 / 865 / 2,358	above	376 / 380 / 412	above
1619	A	3,658	mul_p 334,834, mul_r 1,197,399	39.8 / 1,429 / 5,662	above	39.8 / 150 / 977	above
1619	B	2,993	mul_p 251,513, mul_r 27,761	32.6 / 282 / 1,041	above	32.6 / 52.1 / 199	above
1619	C	2,583	mul_p 207,187, mul_r 1,385	28.1 / 213 / 778	above	28.1 / 42.6 / 152	above

CLASS CHANGES (recorded class vs new class): (‘cover_pushforward’, 211, ‘ic_jc_first_collision’, ‘below’, ‘unresolved (lower bound)’) (‘cover_pushforward’, 211, ‘ic_e1_fibre_first_collision’, ‘below’, ‘unresolved (lower bound)’) (‘cover_pushforward’, 211, ‘birthday_e1_first_collision’, ‘below’, ‘unresolved (lower bound)’) (‘cover_pushforward’, 419, ‘ic_jc_first_collision’, ‘below’, ‘above’) (‘cover_pushforward’, 419, ‘ic_e1_fibre_first_collision’, ‘below’, ‘above’) (‘cover_pushforward’, 419, ‘birthday_e1_first_collision’, ‘below’, ‘unresolved (lower bound)’) (‘cover_pushforward’, 821, ‘ic_jc_first_collision’, ‘below’, ‘above’) (‘cover_pushforward’, 821, ‘ic_e1_fibre_first_collision’, ‘below’, ‘above’) (‘cover_pushforward’, 821, ‘birthday_e1_first_collision’, ‘below’, ‘unresolved (lower bound)’) (‘cover_pushforward’, 1619, ‘ic_jc_first_collision’, ‘below’, ‘above’) (‘cover_pushforward’, 1619, ‘ic_e1_fibre_first_collision’, ‘below’, ‘above’) (‘cover_pushforward’, 1619, ‘birthday_e1_first_collision’, ‘below’, ‘unresolved (lower bound)’) (‘cover_pushforward’, 3203,

‘ic_jc_first_collision’, ‘below’, ‘above’) (‘cover_pushforward’, 3203, ‘ic_e1_fibre_first_collision’, ‘below’, ‘above’) (‘cover_pushforward’, 3203, ‘birthday_e1_first_collision’, ‘below’, ‘above’) (‘tower_large’, 251, ‘g2_recorded’, ‘unresolved’, ‘above’)

Prediction scorecard (frozen in PROTOCOL.md)

id	prediction	result
N1	No arm below the negation rho at all three prices; recorded-arm rows `above` or `unresolved (lower bound)`.	**holds**
N2	Native/step in [1.8, 3.5] (negation), [0.9, 1.6] (plain), four classes.	**refuted**
N3	Negation rho w=1 over w=0 in [1.3, 4]; recorded rho 4 to 10 times the negation total at v3.	**refuted**
N4	Recorded-arm distance up in at least 9 of 11 rows; cover ic_jc new/old in [3, 12].	**holds**
N5	Calibrated order agrees with the band for recorded arms and arm C; mod-p `w_cal` in [0.002, 0.03].	**refuted**
N6	No curve with j = 0 or 1728 among the 32 or more.	**holds**
N7	Arm C (or best of A, B, C) below the recorded arm at all three prices: v3 p = 1619, tower g3 p = 16001, cover ic_jc p = 12841.	**holds**
N8	p = 32003 completes; p = 64007 completes at least one instance.	**holds**
N9	Arm C distance at w = 0 rises with p (rank correlation at least 0.7) in both genera.	**holds**

Details (`scorecard.json`):

- N1: {“below_rows”: [], “recorded_rows_outside”: []}
- N2: negation native/step (four classes) ranges 2.79 to 4.55; plain 1.33 to 2.56; 7 of 20 negation rows and 14 plain rows outside the predicted bands (the per-step overhead of walk starts and DP handling is larger at small r than predicted) [pcc:unverified].
- N3: negation rho w = 1 over w = 0 ranges 1.96 to 7.31 as measured (word operations included) and 1.61 to 2.65 on the four named classes; the second half (recorded rho 4 to 10 times the negation total at v3: 6.83, 5.50, 5.57, 9.22) holds. The first half is scored as measured, so refuted; on the four classes it holds [pcc:unverified].
- N4: 11 of 11 rows up; cover ic_jc new/old at w = 0: 6.88, 4.96, 5.68, 5.32, 7.34, 5.98, 6.28.
- N5: 6 disagreement(s) between band and calibrated class, all on arms with unemitted counters (cover first-collision and birthday arms); the mod-p `w_cal` on the word Jacobian lies in [0.0068, 0.0079]. Scored as frozen (every recorded-arm row), so refuted [pcc:unverified].
- N6: 38 curves checked, 0 with an automorphism beyond negation.

- N7: v3 p = 1619: C totals 2,583 / 23,440 / 211,155 against recorded lower bound 34,521 / 94,893 / 638,242; tower g3 p = 16001: 14,457 / 583,519 / 5,705,076 against 47,918 / 7,242,539 / 71,994,126; cover p = 12841: 15,063 / 229,696 / 2,161,397 against 372,747 / 4,448,291 / 41,128,195 [pcc:unverified].
- N8: tower primes completed [251, 503, 1009, 2003, 4001, 8009, 16001, 32003, 64007].
- N9: rank correlation genus 2 1.00 (n = 9), genus 3 1.00 (n = 9).

Limits

- The unit is one group operation of the arm's own group. A Cantor step costs several elliptic additions (the calibration measured it: `t_add` J / E above), so every IC/rho ratio across group types understates the gap in time [pcc:unverified].
- The recorded v3 and cover collectors do not emit their relation-phase field work; their totals are lower bounds, listed as an implementation gap. Arm L (counted elimination of the same relations) changes the LA count by under 3% (accounting only).
- Cover arms that stop at the first collision emit no native counter; the birthday control and fibre arm cannot be ranked at the calibrated price below the lower-bound line.
- The aut fold was not needed (no curve with j = 0 or 1728); no claim is made for other curves.
- Host timing is best-of-N on a shared VM under variable load; `w_cal` varies by a few tens of percent between runs of this kind and never enters the band verdict.
- No replay certificate or `VALIDATION.json` exists for these rows; they are pre-gate.